



KILLARA HIGH SCHOOL

2010

TRIAL HIGHER SCHOOL CERTIFICATE

EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–8
- All questions are of equal value

Total marks – 120

Attempt Questions 1–8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) $\int_0^{\frac{\pi}{4}} 2 \sec^3 x \tan x \, dx$ 2

(b) Find $\int \frac{dx}{x^2 + 4x + 6}$ 2

(c) Use the substitution $x = 4 \cos^2 \theta$ to evaluate 4

$$\int_0^2 \sqrt{\frac{x}{4-x}} \, dx$$

(d) Use the substitution $t = \tan \frac{x}{2}$ to show that $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{dx}{\sin x} = \ln 3$ 3

(e) (i) Use the substitution $u = \pi - x$ to show that

$$\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{x}{\sin x} \, dx = \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\pi - x}{\sin x} \, dx$$
 3

(ii) Hence find the exact value of $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{x}{\sin x} \, dx$ 1

Question 2 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) Let $z = 3 - 5i$ and $w = 1 - i$. Find zw and $\frac{2}{iw}$ in the form of $x + iy$.

3

- (b) (i) Express $1 + i$ in modulus-argument form.

1

- (ii) Hence evaluate $(1 + i)^{11}$ in the form of $x + iy$.

2

- (c) Sketch the region in the complex number plane where the following inequalities both hold.

$$|z - i| \leq 2 \text{ and } 0 \leq \arg(z + 1) \leq \frac{\pi}{4}.$$

3

- (d) Consider the equation $2z^3 - 3z^2 + 18z + 10 = 0$

- (i) Given that $1 - 3i$ is a root of the equation, explain why $1 + 3i$ is also a root.

1

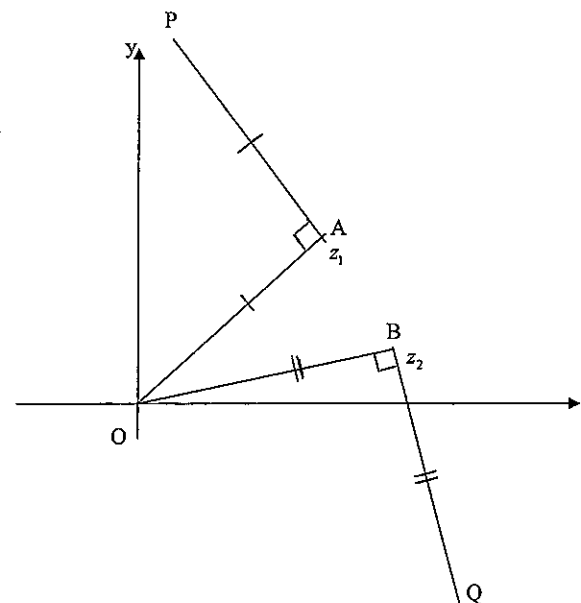
- (ii) Find all the roots of the equation.

1

Question 2 continued

Marks

(e)



The points A and B in the complex number plane correspond to complex numbers z_1 and z_2 respectively. Both triangles OAP and OBQ are right angled isosceles triangles.

- (i) Explain why P corresponds to the complex number $(1 + i)z_1$.

2

- (ii) Let M be the midpoint of PQ. What complex number corresponds to M?

2

End of Question 2

Question 2 continues over the page

Question 3 (15 marks) Use a SEPARATE writing booklet.

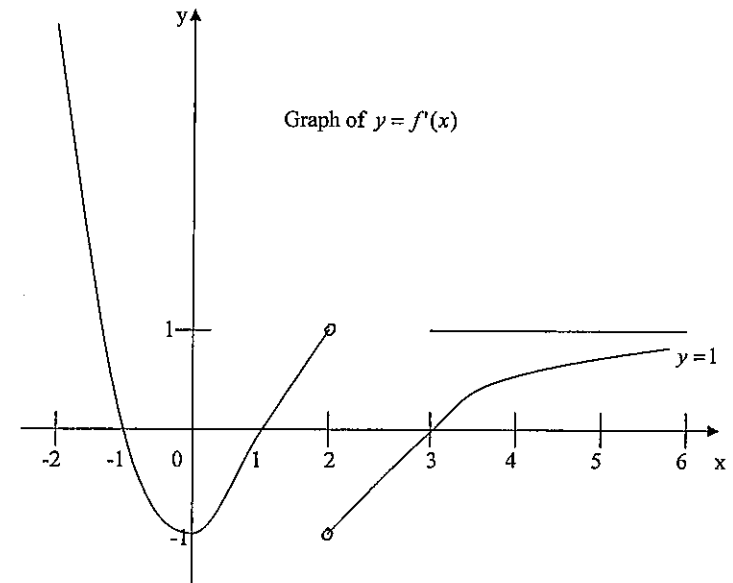
Marks

- (a) Consider the hyperbola $\mathcal{K}$ with equation $xy = 4$.
- (i) Find the points of intersection of $\mathcal{K}$ with the major axis, the eccentricity and the foci of $\mathcal{K}$. 3
 - (ii) Write down the equations of the directrices of $\mathcal{K}$. 1
 - (iii) Sketch $\mathcal{K}$. 2
- (b) Consider the equation $z^3 + mz^2 + nz + 6 = 0$, where m and n are real. It is known that $1 - i$ is a root of the equation.
- (i) Find the other two roots of the equation. 2
 - (ii) Find the values of m and n . 2
- (c) Find the volume of the solid generated by rotating the area bounded by the curve $y = \log_e x$, the x axis and the line $x = 4$. Use the method of cylindrical shells. Rotate the area about the y -axis and give your answer correct to 1 decimal place. 5

Question 4 (15 marks) Use a SEPARATE writing booklet.

Marks

(a)



The diagram shows a sketch of $y = f'(x)$, the derivative function of $y = f(x)$.

The curve $y = f'(x)$ has a horizontal asymptote $y = 1$.

- (i) Identify and classify the turning points of the curve $y = f(x)$. 3
- (ii) Sketch the curve $y = f(x)$ given that $f(0) = 0 = f(2)$ and $y = f(x)$ is continuous. On your diagram, clearly identify and label any important features. 4

Question 4 (continued)

Marks

- (b) The base of a solid is the segment of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. Each cross-section, perpendicular to the major axis of the ellipse, is an equilateral triangle.

(i) Show that the area of the cross-section is $A = y^2\sqrt{3}$. 1

(ii) Hence, or otherwise, find the volume of the solid. 3

- (c) The temperature T_1 of a beaker containing a chemical, and the temperature T_2 of a surrounding vat of cooler water satisfy in accordance with Newton's Law of cooling the equations:

$$\frac{dT_1}{dt} = -k(T_1 - T_2) \text{ and } \frac{dT_2}{dt} = \frac{3}{4}k(T_1 - T_2) \text{ where } k \text{ is a constant.}$$

(i) Show, by differentiation, that $\frac{3}{4}T_1 + T_2 = C$

where C is a constant. 2

(ii) Find an expression for $\frac{dT_1}{dt}$ in terms of T_1 ,

and show that $T_1 = \frac{4}{7}C + Be^{\frac{-7}{4}kt}$ satisfies this differential

equation for any constant B . 2

End of Question 4

Question 5 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) Show that the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point

3

$$P(a \cos \theta, b \sin \theta) \text{ has the equation } \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1.$$

- (ii) This ellipse meets the y-axis at C and D. Tangents drawn at C and D on

3

the ellipse meet the tangent in (i) at the point E and F respectively.

Prove that $CE \cdot DF = a^2$.

- (b) A particle of mass M moves in a straight line with velocity v under the action

of two propelling forces $\frac{Mu^2}{v}$ and Mk^2v where u and k are positive constants.

(i) Show that the acceleration equation $\frac{u^2 + k^2v^2}{v}$ 1

(ii) Show that the distance travelled by the particle in increasing its velocity 4

from $\frac{u}{k}$ to $\frac{2u}{k}$ is $\frac{u}{k^3} \left(1 - \tan^{-1} \frac{1}{3} \right)$.

Question 5 is continued on the next page

Question 5 continued

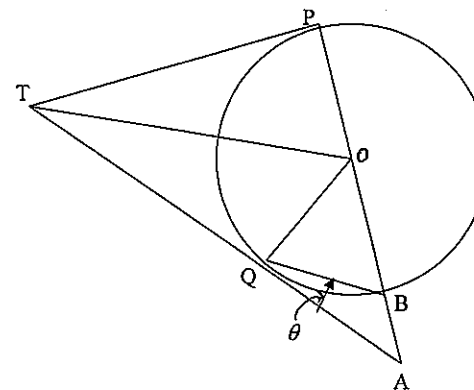
- (c) A code uses a string of the digit 0 and 1 to transmit messages. A message passes through several relay machines each of which sometimes changes the value of an individual digit from 0 to 1 or from 1 to 0. The probability that a digit will be changed by a machine is p .
- (i) Show that the probability that a single digit when received will be different to what was sent after passing through two relay machines is $2p(1 - p)$. 2
- (ii) If there is a 9.5% chance that a digit will be different to what was sent after passing through two relay machines, find the values of p given that p is less than 10 2

End of Question 5

Question 6 (15 marks) Use a SEPARATE writing booklet

Marks

(a)



From an external point T , tangents are drawn to a circle with centre O , touching the circle at P and Q . Angle PTQ is acute.

The diameter PB produced meets the tangent TQ at A . Let $\theta = \angle AQB$.

Copy the diagram above into your answer booklet.

- (i) Prove that $\angle PTQ = 2\theta$. 2
- (ii) Prove that $\triangle PBQ$ and $\triangle TOQ$ are similar. 2
- (iii) Hence show that $BQ \times OT = 2(OP)^2$. 2

Question 6 (continued)

Marks

- (b) If $(x-r)^2$ is a factor of the polynomial $P(x)$, prove that $x-r$ is a factor of the polynomial $P'(x)$.

2

- (c) The polynomial equation $x^4 + x^3 + 1 = 0$ has roots x_1, x_2, x_3 and x_4 .

Construct a polynomial equation whose four roots are x_1^2, x_2^2, x_3^2 and x_4^2 .

3

- (d) The length of an arc joining $P(a,c)$ and $Q(b,d)$ on a smooth continuous curve $y = f(x)$ is given by

$$\text{arc length} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

Consider the curve defined by $y = \frac{x^2}{4} - \frac{\ln x}{2}$.

- (i) Show that $1 + \left(\frac{dy}{dx}\right)^2 = \frac{1}{4} \left(x + \frac{1}{x}\right)^2$.

2

- (ii) Find the length of the arc between $x=1$ and $x=e$.

2

End of Question 6

Question 7 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The equation $ax^3 + bx^2 + d = 0$, has a double root.

5

Show that $27a^2d + 4b^3 = 0$.

- (b) Given that $\sin^{-1} x, \cos^{-1} x$ and $\sin^{-1}(1-x)$ are acute:

5

(i) Show that: $\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1$

(ii) Solve the equation: $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(1-x)$.

- (c) (i) If $\frac{z^2}{z-1}$ is always real, show that the locus of the point represented by z on the argand plane lies on a line and a circle.

4

(i) State which line and which circle.

1

End of Question 7

Question 8 (15 marks) Use a SEPARATE writing booklet

Marks

(a) Let x, y, z and w be positive real numbers.

(i) Prove that $\frac{x}{y} + \frac{y}{x} \geq 2$. 2

(ii) Deduce that $\frac{x+y+z}{w} + \frac{w+y+z}{x} + \frac{w+x+z}{y} + \frac{w+x+y}{z} \geq 12$. 2

(iii) Hence prove that if $x + y + z + w = 1$,

then $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} + \frac{1}{w} \geq 16$. 2

(b) Let $J_n = \int_0^1 x^n e^{-x} dx$, where $n \geq 0$.

(i) Show that $J_0 = 1 - \frac{1}{e}$. 1

(ii) Show that $J_n = nJ_{n-1} - \frac{1}{e}$, for $n \geq 1$. 2

(iii) Show that $J_n \rightarrow 0$ as $n \rightarrow \infty$. 1

(iv) Deduce by the principle of mathematical induction that for all $n \geq 0$,

$$J_n = n! - \frac{n!}{e} \sum_{r=0}^n \frac{1}{r!}. \quad 4$$

(v) Conclude that $e = \lim_{n \rightarrow \infty} \left(\sum_{r=0}^n \frac{1}{r!} \right)$. 1

THE END

Q1 (a) $\int_0^{\frac{\pi}{4}} 2 \sec^3 x \tan x \, dx = \frac{2}{3} [\sec^3 x]_0^{\frac{\pi}{4}} \checkmark$
 $= \frac{2}{3} (2\sqrt{2} - 1) \checkmark$

(b) $\int \frac{dx}{x^2 + 4x + 6} = \int \frac{dx}{(x+2)^2 + 2} = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x+2}{\sqrt{2}} \right) + C \checkmark$

(c) let $x = 4 \cos^2 \theta$ $x=0 \Rightarrow \theta = \frac{\pi}{2}$
 $dx = -8 \cos \theta \sin \theta \, d\theta$ $x=2 \Rightarrow \theta = \frac{\pi}{4}$
 $\int_0^2 \frac{\sqrt{x}}{\sqrt{4-x}} \, dx = - \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \frac{\sqrt{4 \cos^2 \theta}}{\sqrt{4 - 4 \cos^2 \theta}} \cdot 8 \cos \theta \sin \theta \, d\theta \checkmark$
 $= 4 \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \cos 2\theta + 1 \, d\theta \checkmark$
 $= 4 \left[\sin \frac{2\theta}{2} + \theta \right]_{\frac{\pi}{2}}^{\frac{\pi}{4}} \checkmark$
 $= 4 \left(\frac{\pi}{2} - \frac{1}{2} - \frac{\pi}{4} \right)$
 $= \pi - 2 \checkmark$

(d) let $t = \tan \frac{x}{2}$ $x = \frac{2\pi}{3} \Rightarrow t = \sqrt{3}$
 $dt = \frac{1}{2} \sec^2 \frac{x}{2} \, dx$ $x = \frac{\pi}{3} \Rightarrow t = \frac{1}{\sqrt{3}}$
 $\frac{2 dt}{1+t^2} = dx$
 $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{dx}{\sin x} = \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{\frac{x}{t}} \cdot \frac{2 dt}{1+t^2} \checkmark$
 $= \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{dt}{t} = [\ln t]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \checkmark$
 $= \ln \sqrt{3} - \ln \left(\frac{1}{\sqrt{3}} \right) = \ln 3 \checkmark$

(e) let $u = \pi - x$ $x = \frac{2\pi}{3} \Rightarrow u = \frac{\pi}{3}$ and $du = -dx$
 $x = \pi - u$ $x = \frac{\pi}{3} \Rightarrow u = \frac{2\pi}{3}$

$\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{x}{\sin x} \, dx = \int_{\frac{2\pi}{3}}^{\frac{\pi}{3}} \frac{\pi - u}{\sin(\pi - u)} \cdot -du \checkmark$

$= - \int_{\frac{2\pi}{3}}^{\frac{\pi}{3}} \frac{\pi - u}{\sin u} \, du$

$= \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\pi - u}{\sin u} \, du$

$= \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\pi - x}{\sin x} \, dx \checkmark$

$= \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\pi}{\sin x} - \frac{x}{\sin x} \, dx$

$\Rightarrow 2 \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{x}{\sin x} \, dx = \pi \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{dx}{\sin x} \checkmark$
 $= \pi \ln 3$ from (d)

$\therefore \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{x}{\sin x} \, dx = \frac{\pi}{2} \ln 3 \checkmark$

Question 2

a) $\underline{i}$ $z = 3 - 5i$ $w = 1 - i$

$\underline{i}$ $zw = (3 - 5i)(1 - i)$

$$= 3 - 5 - 3i - 5i$$

$$= -2 - 8i \quad \checkmark$$

$\underline{ii}$ $\frac{z}{iw} = \frac{2}{i(1-i)}$

$$= \frac{2}{i+1} \times \frac{1-i}{1-i} \quad \checkmark$$

$$= \frac{2(1-i)}{2}$$

$$= 1 - i \quad \checkmark$$

b) $\underline{i}$ $1 + i$

$$r = \sqrt{2}, \theta = \frac{\pi}{4}$$

$$\therefore 1 + i = \sqrt{2} \operatorname{cis} \frac{\pi}{4} \quad \checkmark$$

$\underline{ii}$ $(1 + i)^{11} = \sqrt{2}^{11} \operatorname{cis} \frac{11\pi}{4}$

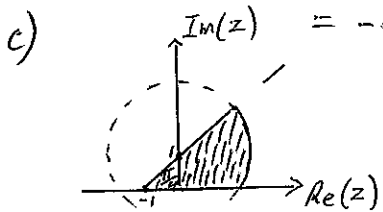
$$= \sqrt{2}^{11} \operatorname{cis} \frac{3\pi}{4} \quad \checkmark$$

$$= \sqrt{2}^{11} \operatorname{cis} \left(-\frac{\pi}{4} \right)$$

$$= \sqrt{2}^{11} \left(\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right)$$

$$= -\sqrt{2}^{10} + i \sqrt{2}^{10} \quad \checkmark$$

$$= -32 + 32i \quad \checkmark$$



- 1 circle
- 1 arg (line)
- 1 shading

d) $\underline{i}$ The complex conjugate $1 + 3i$ of $1 - 3i$ is a root $\checkmark$ because the coefficients of $2z^3 - 3z^2 + 13z + 10 = 0$ are real.

$\underline{ii}$ let α be the 3rd root

$$\therefore \alpha + 1 + 3i + 1 - 3i = -\frac{b}{a} = \frac{3}{2}$$

i.e. $\alpha + \alpha = \frac{3}{2}$

$$\alpha = -\frac{1}{2}$$

$\checkmark \therefore$ the 3 roots are $1 + 3i, 1 - 3i, -\frac{1}{2}$

Question 2

$\underline{i}$ $\vec{OA} = z_1, \vec{AP} = iz_1$

$$\vec{OP} = \vec{OA} + \vec{AP} \quad \checkmark$$

$$= z_1 + iz_1$$

$$= (1 + i)z_1 \quad \checkmark$$

$\underline{ii}$

$$M = \frac{\vec{OP} + \vec{OQ}}{2} \quad \text{where } \vec{OQ} = (1 - i)z_2 \quad \checkmark$$

$$= \frac{z_1(1 + i) + z_2(1 - i)}{2} \quad \checkmark$$

$$\text{or } \frac{z_1 + z_2 + (z_1 - z_2)i}{2}$$

Q3

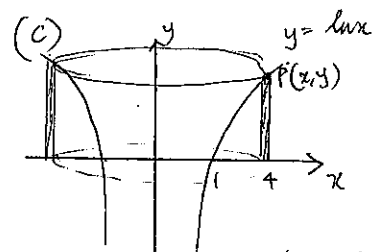
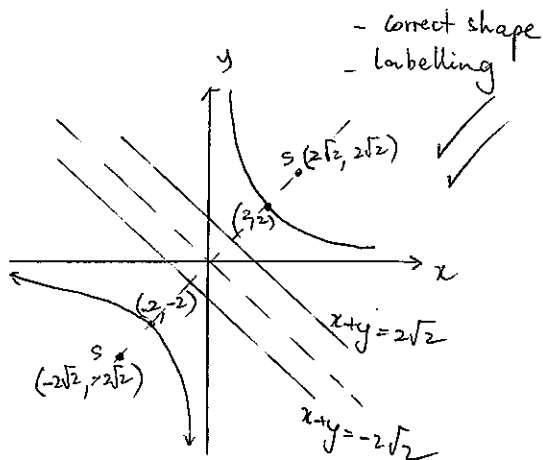
(a) $x \cdot y = 4$
 $= \frac{1}{2} a^2$

$\Rightarrow a = \sqrt{8} = 2\sqrt{2}$

$c^2 = 4 \Rightarrow c = 2$

(i) $(c, c) = (2, 2)$ ✓
 eccentricity $e = \sqrt{2}$ ✓
 foci $(\pm 2\sqrt{2}, \pm 2\sqrt{2})$ ✓

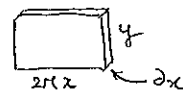
iii) Directrices $x+y = \pm 2\sqrt{2}$ ✓



each cylindrical shell has
 radius = x
 height = y
 thickness = dx

The shell is cut opened to become a rectangle

sheet



$dV = 2\pi x y dx$
 $= 2\pi x \ln x dx$ ✓

Volume of the solid

$$V = \lim_{dx \rightarrow 0} \sum_{x=1}^4 2\pi x \ln x dx$$

$$= 2\pi \int_1^4 x \ln x dx$$

$$= 2\pi \left[\frac{x^2}{2} \ln x \right]_1^4 - \frac{2\pi}{2} \int_1^4 x \cdot \frac{1}{x} dx$$

$$= 2\pi \cdot 8 \ln 4 - \pi \left[\frac{x^2}{2} \right]_1^4$$

$$= 16\pi \ln 4 - \pi \left(\frac{16}{2} - \frac{1}{2} \right)$$

$$= 16\pi \ln 4 - \frac{15\pi}{2}$$

$$= 46.1 \text{ cube units} \checkmark$$

(b) (i) The coeffs' of $P(x)$ are real $\therefore$ complex roots are conjugate
 $\therefore 1+i$ is also a root of $P(x)$ ✓

product of root: $(1+i)(1-i) \cdot \alpha = -6$

$2\alpha = -6$

$\alpha = -3$

$\therefore 1+i$ & 3 are the other 2 roots. ✓

(ii) Sum of roots: $1+i + 1-i + 3 = m$

$\Rightarrow m = 1$ ✓

$\leq$ product of roots
 2 at a time

$(1+i)(1-i) + (1+i)3 + (1-i)3 = n$

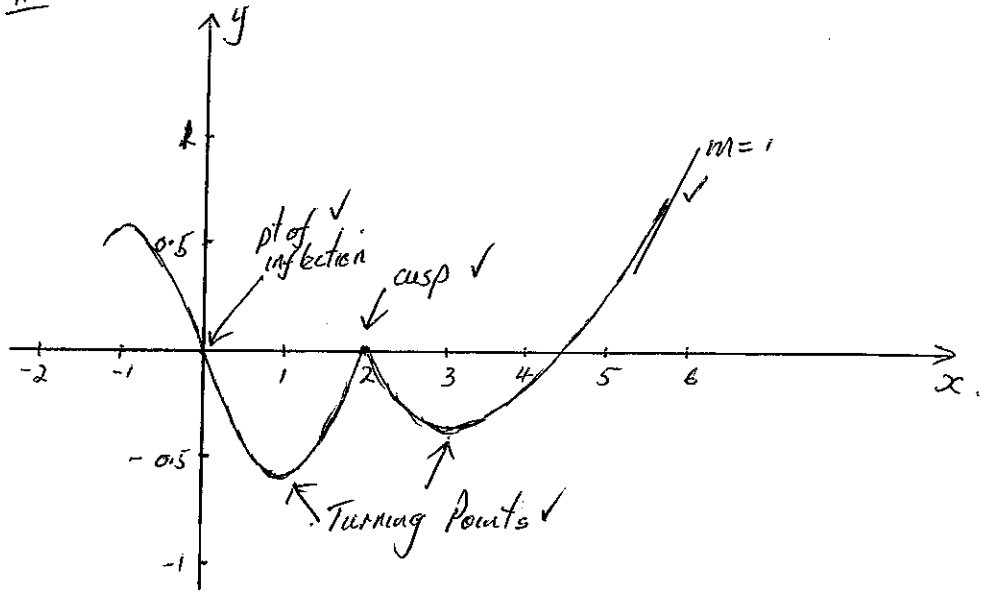
$2 + 3 + 3 = n$

$\therefore n = 8$ ✓

Question 4

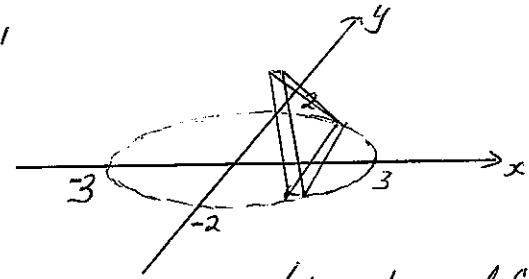
- q i the turning points are at $x = -1, 1$ and 3 ✓
 at $x = -1$, $f'(x)$ is decreasing, $\therefore$ Max T. Pt at $x = -1$
 at $x = 1$, $f'(x)$ is increasing, $\therefore$ Min T. Pt at $x = 1$
 at $x = 3$, $f'(x)$ is increasing, $\therefore$ Min T. Pt at $x = 3$
 3 Marks (1 off for each error).

ii



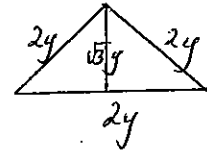
Question 4

b $\frac{x^2}{9} + \frac{y^2}{4} = 1$



i

let the cross section by of " dx " thick with the base " $2y$ " in length.



Area = $\frac{1}{2} \times b \times h = \frac{1}{2} \times 2y \times \sqrt{3}y$ ✓
 $= \sqrt{3}y^2$

ii

Volume = $\lim_{dx \rightarrow 0} \sum \sqrt{3}y^2 dx$
 Volume = $2 \int_0^3 \sqrt{3}y^2 dx$ ✓

$= 2\sqrt{3} \int_0^3 4(1 - \frac{x^2}{9}) dx$

$= 8\sqrt{3} \int_0^3 (1 - \frac{x^2}{9}) dx$ ✓

$= 8\sqrt{3} [x - \frac{x^3}{27}]_0^3$

$= 8\sqrt{3} [3 - 1]$

$= 24\sqrt{3} \text{ units}^3$ ✓
~~16~~

$\frac{x^2}{9} + \frac{y^2}{4} = 1$

$\frac{y^2}{4} = 1 - \frac{x^2}{9}$

$y^2 = 4(1 - \frac{x^2}{9})$

Question 4

c i $\frac{dT_1}{dt} = -k(T_1 - T_2) \quad \frac{dT_2}{dt} = \frac{3}{4}k(T_1 - T_2)$

Consider $\frac{3}{4}T_1 + T_2$

$$\begin{aligned} \frac{d}{dt}\left(\frac{3}{4}T_1 + T_2\right) &= \frac{3}{4}\frac{dT_1}{dt} + \frac{dT_2}{dt} \quad \checkmark \\ &= -\frac{3}{4}k(T_1 - T_2) + \frac{3}{4}k(T_1 - T_2) \\ &= 0 \quad \checkmark \end{aligned}$$

$\therefore \frac{3}{4}T_1 + T_2 = c$ (a constant) — (2)

ii From (2) $T_2 = c - \frac{3}{4}T_1$
Thus $\frac{dT_1}{dt}$ becomes.

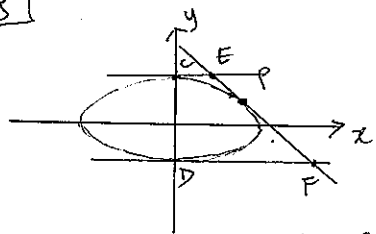
$$\left. \begin{aligned} \frac{dT_1}{dt} &= -k\left(T_1 - c + \frac{3}{4}T_1\right) \quad \checkmark \\ &= kc - \frac{7}{4}kT_1 \end{aligned} \right\}$$

Now Consider $T_1 = \frac{4}{7}c + Be^{-\frac{7}{4}kt}$ — (1)

$$\frac{dT_1}{dt} = 0 - \frac{7kB}{4}e^{-\frac{7}{4}kt} \quad \text{--- (2)}$$

$\checkmark$ but $\frac{dT_1}{dt} = kc - \frac{7}{4}kT_1$
 $= kc - \frac{7}{4}k\left(\frac{4}{7}c + Be^{-\frac{7}{4}kt}\right)$ from (1)
 $= kc - kc - \frac{7kB}{4}e^{-\frac{7}{4}kt}$
 $= 0 - \frac{7kB}{4}e^{-\frac{7}{4}kt}$ which is
 equal to the above result (2)
 $\therefore T_1 = \frac{4}{7}c + Be^{-\frac{7}{4}kt}$ satisfies $\frac{dT_1}{dt}$.

Q5



gradient of the tangent

$$\frac{2x}{a^2} dx + \frac{2y}{b^2} dy = 0$$

$$\Rightarrow m = -\frac{b^2 x}{a^2 y}$$

$$\text{at } P(a \cos \theta, b \sin \theta) \Rightarrow m = -\frac{b^2 \cdot a \cos \theta}{a^2 b \sin \theta} = -\frac{b \cos \theta}{a \sin \theta}$$

equation of the tangent at P

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$\frac{y \sin \theta}{b} + \frac{x \cos \theta}{a} = \sin^2 \theta + \cos^2 \theta$$

$$\Rightarrow \frac{y \sin \theta}{b} + \frac{x \cos \theta}{a} = 1 \quad (1)$$

$\therefore \frac{y \sin \theta}{b} + \frac{x \cos \theta}{a} = 1$ is the equation of the tangent at P.

ii) equation of the tangent at C is $y = b$

equation of the tangent at D is $y = -b$

Sub. $y = \pm b$ into (1)

$$\frac{b \sin \theta}{b} + \frac{x \cos \theta}{a} = 1 \quad \text{and} \quad \frac{-b \sin \theta}{b} + \frac{x \cos \theta}{a} = 1$$

$$x = \frac{a(1 - \sin \theta)}{\cos \theta}$$

$$x = \frac{a(1 + \sin \theta)}{\cos \theta}$$

coordinates of E $\left(b, \frac{a(1 - \sin \theta)}{\cos \theta}\right)$ and F $\left(\frac{a(1 + \sin \theta)}{\cos \theta}, -b\right)$

$$CE \cdot DF = \frac{a(1 - \sin \theta)}{\cos \theta} \cdot \frac{a(1 + \sin \theta)}{\cos \theta} = \frac{a^2(1 - \sin^2 \theta)}{\cos^2 \theta} = a^2$$

(b) The force acts on the particle is:

$$ma = \left(\frac{Mu^2}{v} + Mh^2 v\right)$$

$$= M \left(\frac{u^2}{v} + v h^2\right)$$

$$\therefore a = \frac{u^2 + v^2 h^2}{v}$$

$\therefore$ The equation of motion is

$$\ddot{x} = \frac{u^2 + v^2 h^2}{v}$$

$$v \frac{dv}{dx} = \frac{u^2 + v^2 h^2}{v}$$

$$\frac{dv}{dx} = \frac{u^2 + v^2 h^2}{v^2} \Rightarrow dx = \frac{v^2}{u^2 + v^2 h^2} dv$$

$$\therefore h^2 dx = \left(1 - \frac{u^2}{u^2 + v^2 h^2}\right) dv$$

integrate both sides

$$\Rightarrow h^2 x = v - \frac{u}{h} \tan^{-1} \frac{h v}{u} + C$$

$$\text{when } x=0 \quad v = \frac{u}{h} \Rightarrow C = \frac{u}{h} \tan^{-1} 1 - \frac{u}{h}$$

$$\therefore x = \frac{v}{h^2} - \frac{u}{h^3} \tan^{-1} \frac{h v}{u} + \frac{u}{h^3} \tan^{-1} 1 - \frac{u}{h^3}$$

$$\text{when } v = \frac{2u}{h}$$

$$x = \frac{2u}{h^3} - \frac{u}{h^3} + \frac{u}{h^3} \left(\tan^{-1} 2 - \tan^{-1} 1\right) = \frac{u}{h^3} \left(1 + \tan^{-1} \frac{1}{3}\right)$$

Note:

$$\text{let } A = \tan^{-1} 2, B = \tan^{-1} 1$$

$$\tan(A - B) = \frac{2-1}{1+2 \times 1} = \frac{1}{3}$$

$$\therefore A - B = \tan^{-1} \frac{1}{3}$$

$$\text{hence } \tan^{-1} 2 - \tan^{-1} 1 = \tan^{-1} \frac{1}{3}$$

(c) Q5 Cont

$$P(\text{digit will be changed}) = p$$

$$P(\text{digit won't be changed}) = 1-p$$

$$\begin{aligned} \text{(i)} \quad P &= p(1-p) + (1-p) \cdot P \\ &= 2p(1-p) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{ii)} \quad 2p(1-p) &= 0.095 \\ 2p^2 - 2p + 0.095 &= 0 \quad \checkmark \\ p &= \frac{2 \pm \sqrt{4 - 4 \times 2 \times 0.095}}{4} \\ &= \frac{2 \pm 1.8}{4} \end{aligned}$$

$$\Rightarrow p = 95\% \quad \text{or} \quad 5\% \quad \checkmark$$

Since p is less than 10%, $p = 5\%$. $\checkmark$

Question 6

a i $\angle BQA = \theta$
 $\therefore \angle BPQ = \theta$ (alternate segment theorem) $\checkmark$
 $\therefore \angle PQO = \theta$ ($\triangle OQP$ is isosceles)
 $\therefore \angle POQ = 180 - 2\theta$ (angle sum of $\triangle POQ$)
 Now $\angle TPO = \angle + QO = 90^\circ$ (radius and tangent) $\checkmark$
 $\therefore \angle PTQ = 2\theta$ (angle sum of quadrilateral $TPOQ$)

ii Consider $\triangle PBQ$ and $\triangle TOQ$
 $\ast \angle PQB = 90$ (angle in a semi circle)
 $= \angle TQO$ $\checkmark$

$\ast \angle OTQ = \theta$
 $= \angle QPO$ ($\angle QPB$) from i $\checkmark$

Hence $\triangle PBQ \parallel \triangle TOQ$ (AA).

iii $\frac{PB}{TO} = \frac{BQ}{OQ}$ (corresponding sides of similar triangles)
 However, $PB = 2OP$ and $OQ = OP$ $\checkmark$

$\therefore \frac{2OP}{TO} = \frac{BQ}{OP}$ $\checkmark$

Thus $BQ \times OT = 2(OP)^2$

b let $P(x) = (x-2)^2 \cdot Q(x)$ where $Q(x)$ is a polynomial
 $\therefore P'(x) = 2(x-2) \cdot Q(x) + (x-2)^2 \cdot Q'(x)$
 $= (x-2)[2Q(x) + (x-2)Q'(x)]$ $\checkmark$
 $\therefore x-2$ is a factor of $Q(x)$.

Question 6

c Replace x with $\sqrt{x}$
 $\therefore (\sqrt{x})^4 + (\sqrt{x})^3 + 1 = 0$ $\checkmark$
 $x^2 + x^{\frac{3}{2}} + 1 = 0$
 $x^{\frac{3}{2}} = -(x^2 + 1)$
 $x^3 = (x^2 + 1)^2$ $\checkmark$
 $x^3 = x^4 + 2x^2 + 1$
 $x^4 - x^3 + 2x^2 + 1 = 0$ $\checkmark$

d i
 $y = \frac{x^2}{4} - \frac{\ln x}{2}$
 $\frac{dy}{dx} = \frac{x}{2} - \frac{1}{2x}$
 $= \frac{1}{2}(x - \frac{1}{x})$ $\checkmark$

Now, $1 + (\frac{dy}{dx})^2 = 1 + \frac{1}{4}(x - \frac{1}{x})^2$
 $= 1 + \frac{1}{4}(x^2 + \frac{1}{x^2} - 2)$
 $= \frac{1}{4}(x^2 + \frac{1}{x^2} + 2)$
 $= \frac{1}{4}(x + \frac{1}{x})^2$ $\checkmark$

ii Arc length $= \int_a^b \sqrt{1 + (\frac{dy}{dx})^2} dx$
 $= \int_1^e \sqrt{\frac{1}{4}(x + \frac{1}{x})^2} dx$ $\checkmark$
 $= \frac{1}{2} \int_1^e (x + \frac{1}{x}) dx$
 $= \frac{1}{2} \left[\frac{x^2}{2} + \ln x \right]_1^e$
 $= \frac{1}{2} \left(\frac{e^2}{2} + 1 - \frac{1}{2} \right)$
 $= \frac{1}{2} \left(\frac{e^2}{2} + \frac{1}{2} \right)$
 $= \frac{1}{4} (e^2 + 1)$ units $\checkmark$

(Q7)

$$(a) \quad ax^3 + bx^2 + d = 0$$

$$P'(x) = 3ax + 2bx \quad \checkmark$$

$$P''(x) = 6a + 2b$$

for double root $P'(x) = 0$

$$\therefore x(3ax + 2b) = 0$$

$$x = 0 \text{ or } x = -\frac{2b}{3a} \quad \checkmark$$

* if $x=0$ is a double root then $P(0) = d = 0$

and if $27a^2d + 4b^3 = 0$ then $4b^3 = 0 \Rightarrow b = 0$

$\therefore P(x) = ax^3$ hence 0 is a tripple root of $P(x)$ and

$$27a^2d + 4b^3 \neq 0 \quad \checkmark$$

* if $x = -\frac{2b}{3a}$ is a double root then

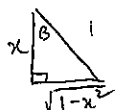
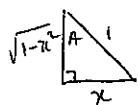
$$P\left(-\frac{2b}{3a}\right) = -a\left(\frac{2b}{3a}\right)^3 + b\left(\frac{2b}{3a}\right)^2 + d = 0$$

$$-\frac{8b^3}{27a^2} + \frac{4b^3}{9a^2} + d = 0 \quad \checkmark$$

$$\Rightarrow \frac{4b^3 + 27a^2d}{27a^2} = 0 \quad \checkmark$$

$$\therefore 27a^2d + 4b^3 = 0$$

(b) let $A = \sin^{-1}x$, $B = \cos^{-1}x$



$$(i) \quad \sin(\sin^{-1}x - \cos^{-1}x) = \sin(A - B)$$

$$= \sin A \cos B - \cos A \sin B$$

$$= x \cdot x - \sqrt{1-x^2} \cdot \sqrt{1-x^2} \quad \checkmark$$

$$= x^2 - 1 + x^2$$

$$= 2x^2 - 1 \quad \checkmark$$

ii)

$$\sin^{-1}x - \cos^{-1}x = \sin^{-1}(1-x)$$

$$\sin(\sin^{-1}x - \cos^{-1}x) = \sin(\sin^{-1}(1-x))$$

$$2x^2 - 1 = 1 - x \quad \text{from (i)} \quad \checkmark$$

$$2x^2 + x - 2 = 0$$

$$x = \frac{-1 \pm \sqrt{1+4 \times 2 \times 2}}{4}$$

$$= \frac{-1 \pm \sqrt{17}}{4} \quad \checkmark$$

$$\text{as } -1 \leq x \leq 1, \quad x = \frac{-1 + \sqrt{17}}{4} \quad \checkmark$$

(c) if $\frac{z^2}{z-1}$ is always real then $\text{Im}(z) = 0$

$$\text{let } z = a + ib$$

$$\frac{z^2}{z-1} = \frac{(a+ib)^2}{(a-1)+ib}$$

$$= \frac{(a^2 - b^2 + 2abi)(a-1-ib)}{(a-1)+ib(a-1-ib)}$$

$$= \frac{(a^2 - b^2)(a-1) + 2ab^2 + (a-1)2abi - (a^2 - b^2)ib}{(a-1)^2 + b^2} \quad \checkmark$$

$$\text{Im } z = \frac{(a-1)2ab - (a^2 - b^2)b}{(a-1)^2 + b^2} = 0 \quad \checkmark$$

$$\Rightarrow b(2a^2 - 2a - a^2 + b^2) = 0$$

$$b(a^2 - 2a + 1 + b^2 - 1) = 0 \quad \checkmark$$

$$b((a-1)^2 + b^2 - 1) = 0$$

$$\therefore b = 0 \quad \text{or} \quad (a-1)^2 + b^2 = 1$$

$$\text{let } x = a \text{ and } y = b$$

$$\text{Then } y = 0 \text{ and } (x-1)^2 + y^2 = 1 \quad \checkmark$$

ii) The line is the x-axis except $x=1$ and the circle has radius = 1 unit & the centre (1,0)

Question 8

i $\left(\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}}\right)^2 = \frac{x}{y} + \frac{y}{x} + 2 \quad \checkmark$

But $\left(\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}}\right)^2 \geq 0 \quad \checkmark$

$\therefore \frac{x}{y} + \frac{y}{x} + 2 \geq 0$

$\frac{x}{y} + \frac{y}{x} \geq -2$

ii $\frac{x+y+z}{w} + \frac{w+y+z}{x} + \frac{w+x+z}{y} + \frac{w+x+y}{z} \quad \text{--- ①}$

$= \frac{x}{w} + \frac{y}{w} + \frac{z}{w} + \frac{w}{x} + \frac{y}{x} + \frac{z}{x} + \frac{w}{y} + \frac{x}{y} + \frac{z}{y} + \frac{w}{z} + \frac{x}{z} + \frac{y}{z}$

$= \left(\frac{x}{w} + \frac{w}{x}\right) + \left(\frac{y}{w} + \frac{w}{y}\right) + \left(\frac{z}{w} + \frac{w}{z}\right) + \left(\frac{y}{x} + \frac{x}{y}\right) + \left(\frac{z}{x} + \frac{x}{z}\right) + \left(\frac{z}{y} + \frac{y}{z}\right) \quad \checkmark$

$\geq 2 + 2 + 2 + 2 + 2 + 2 = 12 \quad (\text{from i}) \quad \checkmark$

iii $x+y+z+w=1$

$\therefore x+y+w=1-z, x+y+z=1-w, x+w+z=1-y, w+y+z=1-x$

substitute into ① in ii

$\therefore \frac{1-w}{w} + \frac{1-x}{x} + \frac{1-y}{y} + \frac{1-z}{z} \geq 12 \quad \checkmark$

$\frac{1}{w} + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} - 4 \geq 12$

$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} + \frac{1}{w} \geq 16 \quad \checkmark$

Question 8

i $J_0 = \int_0^1 e^{-x} dx$

$= -[e^{-x}]_0^1$

$= -(e^{-1} - e^0) \quad \checkmark$

$= 1 - \frac{1}{e}$

ii let $u = x^n$ and $v = -e^{-x}$
 $u' = nx^{n-1}$ and $v' = e^{-x}$

$\therefore J_n = \int_0^1 x^n e^{-x} dx$

$= \int_0^1 uv' dx$

Now $\int_0^1 uv' dx = uv - \int v u' dx$

$\int_0^1 x^n e^{-x} dx = -[x^n e^{-x}]_0^1 - \int_0^1 nx^{n-1}(-e^{-x}) dx \quad \checkmark$

$\therefore 0 \leq \int_0^1 x^n e^{-x} dx \leq \int_0^1 x^n dx$

$J_n = -e^{-1} + n \int_0^1 x^{n-1} e^{-x} dx$

However, $\int_0^1 x^n dx = \frac{1}{n+1} \rightarrow 0$ as $n \rightarrow \infty$

$= -e^{-1} + n J_{n-1} \quad \checkmark$

$\therefore \int_0^1 x^n e^{-x} dx \rightarrow 0$ as $n \rightarrow \infty \quad \checkmark$

$\therefore J_n \rightarrow 0$ as $n \rightarrow \infty$

✓ From IV (PTD)

$J_n = n! - \frac{n!}{e} \sum_{r=0}^n \frac{1}{r!}$

$\therefore \sum_{r=0}^n \frac{1}{r!} = -\frac{e J_n}{n!} + e$

as $n \rightarrow \infty$, RHS $\rightarrow 0 + e$ from (iii)

Thus $\lim_{n \rightarrow \infty} \left(\sum_{r=0}^n \frac{1}{r!} \right) = e \quad \checkmark$

Question 8

Δ iv * When $n=0$, $J_0 = 0! - \frac{0!}{e} \sum_{r=0}^0 \frac{1}{r!}$
 $= 1 - \frac{1}{e}$

$\therefore$ true for $n=0$ ✓

* Assume true for $n=k$, $k > 0$

i.e. $J_k = k! - \frac{k!}{e} \sum_{r=0}^k \frac{1}{r!}$

* We need to prove true for $n=k+1$

i.e. $J_{k+1} = (k+1)! - \frac{(k+1)!}{e} \sum_{r=0}^{k+1} \frac{1}{r!}$?

* LHS = J_{k+1}

$= (k+1)J_k - \frac{1}{e}$ ✓ from (i)

$= (k+1) \left[k! - \frac{k!}{e} \sum_{r=0}^k \frac{1}{r!} \right] - \frac{1}{e}$

$= (k+1) \frac{k!}{e} - \frac{(k+1)!}{e} \sum_{r=0}^k \frac{1}{r!} - \frac{1}{e}$

$= (k+1)! - \frac{(k+1)!}{e} \sum_{r=0}^k \frac{1}{r!} - \frac{(k+1)!}{e} \left(\frac{1}{(k+1)!} \right)$

$= (k+1)! - \frac{(k+1)!}{e} \left[\sum_{r=0}^{k+1} \frac{1}{r!} - \frac{1}{(k+1)!} \right]$ ✓

$= (k+1)! - \frac{(k+1)!}{e} \sum_{r=0}^{k+1} \frac{1}{r!}$ (required form)

* Thus, if true for $n=k$ then true for $n=k+1$
 It is true for $n=0$, $\therefore$ true for $n=1, 2, 3$
 and so on. Thus true for all $n \geq 0$.